



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1

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❖ **Goal:** Set up your environment and write your first C program.

➤ **Install a C Compiler / IDE →** You need a place to write and run your code. Choose one:

- **Windows:** Code::Blocks, Dev C, VS Code with C extension
- **Mac/Linux:** VS Code or terminal with g++ installed
- **Online (no install):** Replit, OnlineGDB, GeeksforGeeks IDE
- **Android :** Termux, Cxxdroid, Coding C

❖ **Use IDE – Codeblock**



Code::Blocks

1. Step 1.

Download Link -

<https://sourceforge.net/projects/codeblocks/files/Binaries/25.03/Windows/codeblocks-25.03mingw-setup.exe>



Microsoft Windows (64 bit, default)

File	Download from
codeblocks-25.03-setup.exe	Sourceforge.net or dAppCDN.com
codeblocks-25.03-setup-nonadmin.exe	Sourceforge.net or dAppCDN.com
codeblocks-25.03-nosetup.zip	Sourceforge.net or dAppCDN.com
codeblocks-25.03mingw-setup.exe	Sourceforge.net or dAppCDN.com
codeblocks-25.03mingw-nosetup.zip	Sourceforge.net or dAppCDN.com

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2. Step 2.

Click → codeblocks-25.03mingw-setup.exe → Sourceforge.net

3. Step3.

Automatic Download and install carefully.

❖ Use IDE – VS Code



Visual Studio Code

- Download VS Code → <https://code.visualstudio.com/download>
- VS Code does not have a compiler, it is only a Code editor. Step 1 download any compiler e.g. – GCC/g++/MinGW, clog,.....

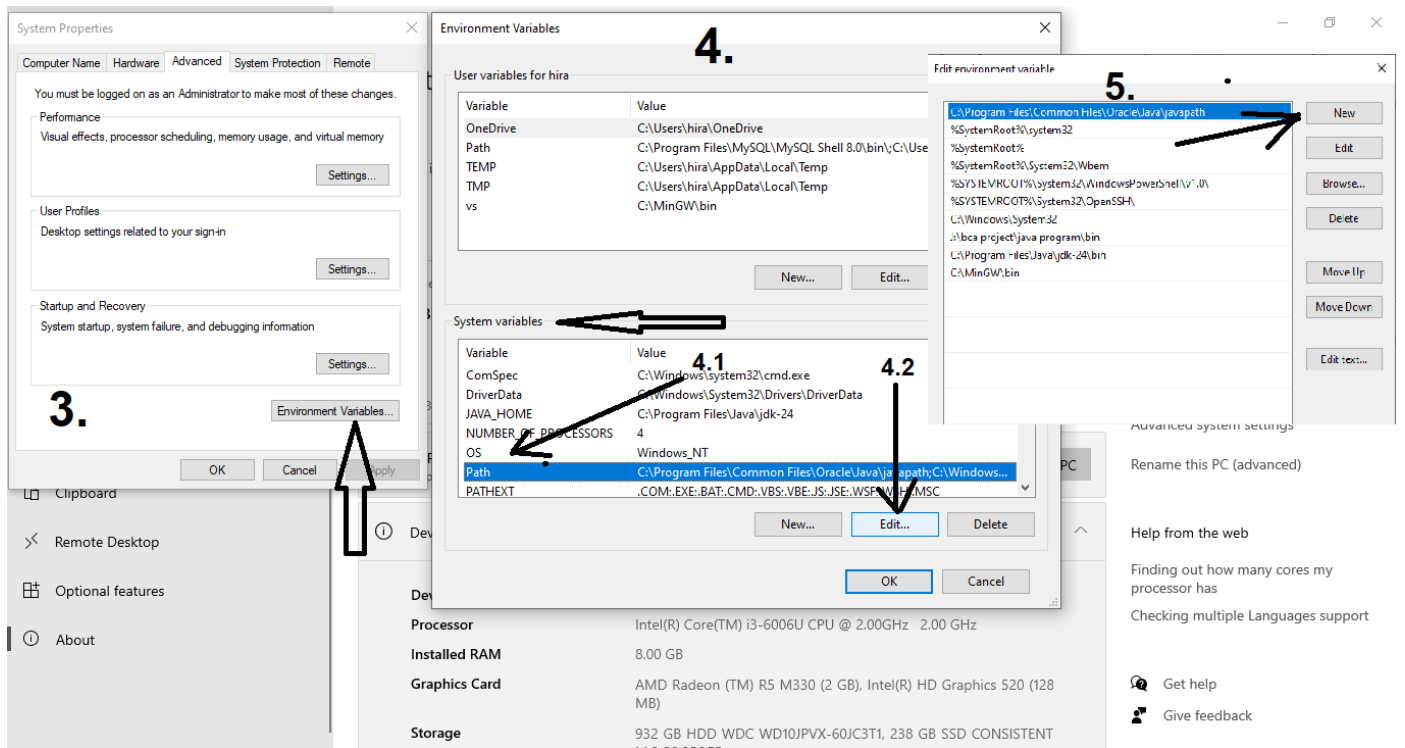
Step1 : Website → <https://www.msys2.org/>

Direct Download link → https://github.com/msys2/msys2-installer/releases/download/2025-08-30/msys2-x86_64-20250830.exe

Install Location keep on → c:\mingw64\bin

Step 2: Set Path

1. This PC → Right click → Properties (Open setting window)
2. In Setting → About → Advanced system settings
3. Choose → Environment Variables
4. In System Variables → Choose Path → and click Edit button
5. New button click and type –
C:\mingw64\bin
6. Click → ok → ok → PC restart



Step 3: Check Compiler installed or Not

1. Open Command prompt (cmd)
2. Type → `g++ --version`
3. See version then Compiler have been installed. Other wise Check again all steps.

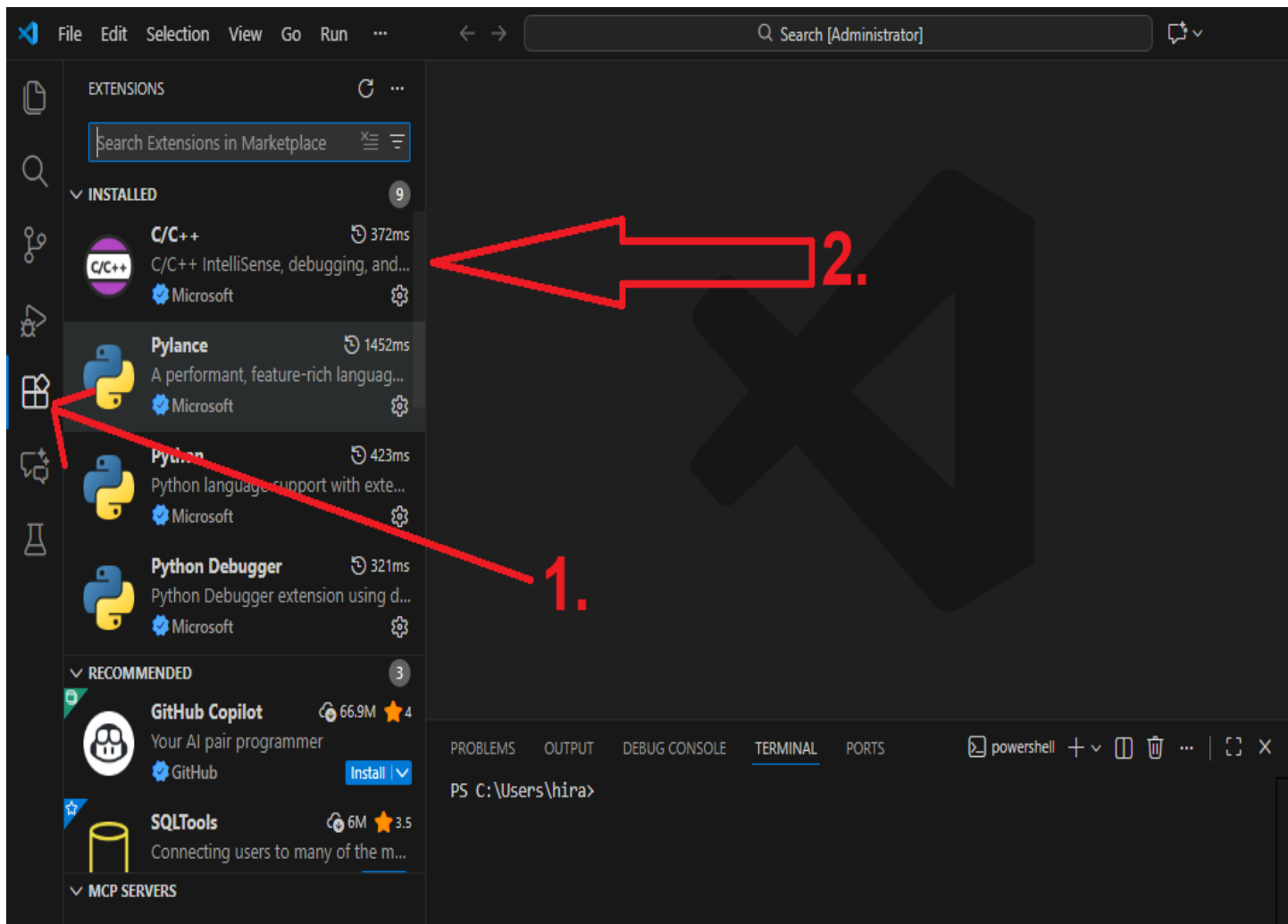
```
C:\>g++ --version
g++ (GCC) 13.2.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

C:\>gcc --version
gcc (GCC) 13.2.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Now –

Step 4 : C extension Install in VS Code

1. Open VS Code & Left Side Extension icon OR (Ctrl + Shift + X)
2. Search → c/c
3. Install Microsoft version extension



❖ Let's Start programming in C

Q1. Print "Hello, World !" In C language.

Structure of Basic C Program

Sl.No	Code	What is this
1.	#include<stdio.h>	Preprocessor Directive
2.	int main()	Main function
3.	{	Entry Point
4.	Printf("Hello, world ! ");	Output statement
5.	Return 0;	Finish/Exit program,return value 0 (0 means "Success") नये compiler return 0; optional हैं, compiler automatic return 0; मान लेती हैं
6.	}	Exit Point

Understand Each Line

SI No	Code	What is work
1.	#include<stdio.h>	Include input/output functions like printf(),scanf()
2.	int main()	Every C program starts from here
3.	{	Entry point
4.	printf("Hello, world ! ");	Used to display output
5.	Return 0;	Ends the program successfully.
6.	}	Exit point

Screen View – Source Code with Output

- **\n = Escape sequence** = Use New line **OR** line change one character, but don't flush buffer memory. It's Performance is fast
- **\t** = 1 tab means 4 space

❖ Tokens

- A token is the smallest element of a C program that is meaningful to the compiler. It act as building blocks of a program.
- The following tokens are available in C keywords, identifiers, constants, variables, and operators described in this section are examples of tokens. Punctuation characters such as brackets ([]), braces ({ }), parentheses (()), and commas (,) are also tokens.

➤ Name of tokens

- ✓ Keywords
- ✓ Identifiers
- ✓ Constants
- ✓ Variables
- ✓ Operators

✓ Punctuation

❖ Variables

- In C a variable is a place to store information. A variable is a location in your computer's memory in which you can store a value and from which you can later retrieve that value.
- Data value is Change or may change due to some calculation during the program execution.
- When you define a variable in C, you must tell the compiler what kind of variable it is: an integer, a character, and so forth.
- When you run your program, it is loaded into RAM from the disk file. All variables are also created in RAM.

Declaration of Variable

Declaration will allocate memory for specified variable with garbage value.

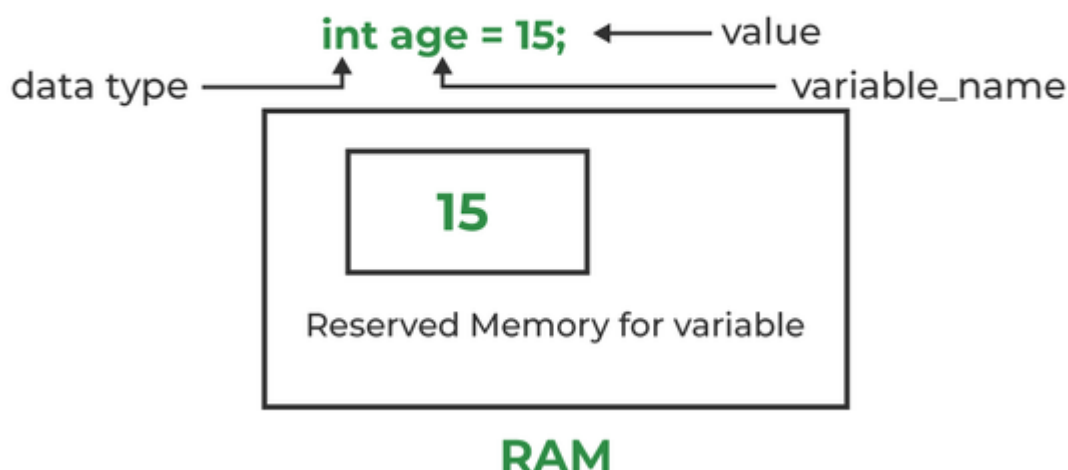
Syntax :

```
Data-Type    Variable-name;
```

Examples :

```
int a;  
float b;  
char c;
```

Variable in C++



❖ Initialization of Variable in Program

Initialization means assigning value to declared variable. Every value will overwrite the previous value.

Q1. Write a C program to take the username, Age, mobile number and salary from the user.

Q2. Write a program to print your 10th class marks in percentage.

❖ C Identifiers

In C programming language, identifiers are the unique names assigned to variables, functions, classes, structs, or other entities within the program.

➤ Rule = How to write Variable_name / Identifiers

- It must begin with alphabet or underscore. (e.g – a2, Hira, _2, bihar_sharif,)
- No other symbol is allowed (e.g – a\$, a.b, x-1,...)
- 1st variable name cannot start with a digit (e.g – 2a, 8x,...)
- C has reserved keywords that cannot be used as identifiers.
- Identifier must be unique in its namespace.

Valid Identifiers	Invalid Identifiers
• Hira; • A; • num; • bihar_sharif; • a2; • sum;	• 1hira (start with digit) • bihar sharif (space not allow) • goto (keyword) • 2a; (start with digit) • Bihar@123 (special symbol not allow)

❖ Keywords

- It is also called Pre-define words / Reserved word /type of token
- Keywords (reserved words) have special meanings to the C compiler and are always written or typed in lower cases.
- Keywords are words that the language uses for a special purpose, such as void, int, public, etc.
- It can't be used for a variable name or function name or any other identifiers.
- The total 32 keywords present in C language, but now 44 keywords available.
- Categorization of c keywords ↵

asm	dynamic_cast	new	template
auto	else	operator	this
bool	enum	private	throw
break	extern	protected	true
case	false	public	try
catch	float	register	typedef
char	for	reinterpret_cast	typeid
class	friend	return	union
const	goto	short	unsigned
const_cast	if	signed	using
continue	inline	sizeof	virtual
default	int	static	void
delete	long	static_cast	volatile
do	mutable	struct	wchar_t
double	namespace	switch	while

❖ Constants

- Constants (often referred to as Literals) are data items that never change their value during the execution of the program.

Type of Constants

The following types of constants are available in C.

- ✓ Integer Constants
- ✓ Character Constants
- ✓ Floating Point Constants
- ✓ Strings Constants
- **Integer constant**-Consists of the digits.
 - ✓ Example :- 10, -555, 10A(hexa form), 22, -9,
- **Float constant**-Consists of the digits with decimal.
 - ✓ Example :- 3.0, -7.5, -0.067, 45.55, 2.5e3,
- **Character constant**-single character enclose with in a part of single quote.
 - ✓ Example :- 'A', '9', '#',
- **String constant**- Collection is sequence of characters enclosing in double quotes.
 - ✓ Example :- "Hira", "New Delhi",

❖ C Modifiers

- Modifiers are used with primary data types like int, float, char to control size, sign, or behavior.

Data type modifiers available in C are:

- ✓ Signed → Can store positive & negative values
- ✓ Unsigned → Can store only positive values
- ✓ Short → Reduces the size of integer type
- ✓ Long → increases the size of integer type

Modifiers in C++



Old 16bit /Turbo C++ sizes

Type	Size (bits)	Size (bytes)	Range
char	8	1	-128 to 127
unsigned char	8	1	0 to 255
int	16	2	-2^{15} to $2^{15}-1$
unsigned int	16	2	0 to $2^{16}-1$
short int	8	1	-128 to 127
unsigned short int	8	1	0 to 255
long int	32	4	-2^{31} to $2^{31}-1$
unsigned long int	32	4	0 to $2^{32}-1$
float	32	4	3.4E-38 to 3.4E+38
double	64	8	1.7E-308 to 1.7E+308
long double	80	10	3.4E-4932 to 1.1E+4932

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